जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

18th July 2025

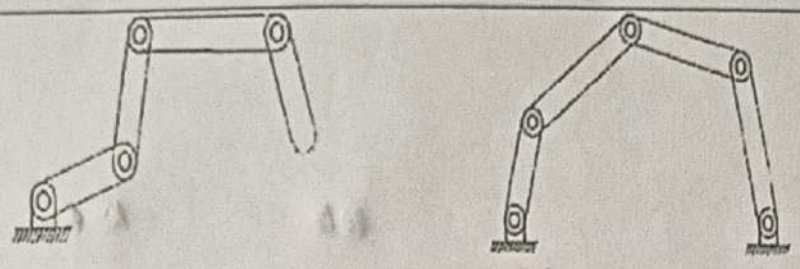
Roll No. [redacted] B E C S E [redacted]

SEMESTER MAJOR EXAMINATIONS

Programme	:	B.Tech CSE / B.Tech CSE (Cyber Security)	Semester	:	6th
Course Name	:	Robotics and AI	Course Code	:	BECCS3C009 / BECSE3C009
Duration	:	3 Hours	Credits	:	03
Maximum Marks	:	75	Minimum Marks	:	

After pursuing this course, the student will be able to:	
CO 01	Learn the fundamental principles, components, and applications of robotics and AI.
CO 02	Understand the designing methodology of integrating actuators, sensors, and control mechanisms in robotic systems.
CO 03	Explore AI techniques such as NLP, computer vision, and reinforcement learning for robotic systems
CO 04	Investigate tools like ROS and Gazebo for robot simulation and assess IoT integration in robotics.
CO 05	Explore robotic process automation (RPA) tools and their applications in various domains.

Part A Very Short Answer Type Logical Questions of 03 marks related to different COs (All the units to be covered from the entire syllabus)		Total Marks 3X5=15	
S.No.	Statement	Levels	CO
1.	Discuss the concept of locomotion and its variants, and analyze their applicability in robotic systems use cases.	L4	CO-01, CO-02
2.	Highlight the key mechanism of Publish-Subscribe, Push-Pull, and Exclusive Pair communication models?	L3	CO-01
3.	Discuss various stages of Natural Language Processing?	L1	CO-03
4.	How is co-ordinative multi-robot system different from a competitive multi-robot system? Provide real-world examples for each category.	L3	CO-03
5.	What is Robotic Process Automation (RPA), and how does it work in the context of UiPath?	L2	CO-05
PART B Short Answer Type Analytical Questions of 06 marks related to different COs (All the units to be covered from the entire syllabus)		Total Marks 6X5=30	
6.	Analyse the working mechanism of following control systems with reference to robotic systems: i) Open-Loop Control ii) Closed-Loop Control iii) Feed-Forward Control iv) Feedback Control	L4	CO-02
7.	Explain the mechanism of image recognition using Convolutional Neural Network (CNN)?	L2	CO-03
8.	How does the Degree of Freedom (DOF) impact system flexibility? Evaluate the DOF of the following systems:		

	 (a) (b)	L5	CO-01, CO-02
9.	Explain the following metrics used in the evaluation of Speech Recognition in NLP systems, along with their formulas: a) Word Error Rate (WER) b) Character Error Rate (CER) c) Match Error Rate (MER)	L2	CO-03
10.	What types of tasks or processes can be automated using UiPath, and how does UiPath facilitate this automation? Also, provide some examples of the same.	L3	CO-05
Part C **Long Answer Type Descriptive Questions of 15 marks related to different Cos** **(Only Two questions to be attempted)**		**Total Marks 15X2=30**	
11.	i) Discuss the key components of robotic systems and explain their roles in system functionality. (8 marks) ii) Compare and contrast various types of intelligent agents, providing use cases for each.	L2, L4	CO-01, CO-03
12.	i) Provide a detailed discussion on the architecture of ROS (Robot Operating System). (8 marks) ii) Explain the mechanism of MQTT protocols, along with an example. (7 marks)	L2	CO-04
13.	Discuss the following modules with reference to Human-Robot Interaction (HRI) along with depictions: a) Automatic Speech Recognition (ASR) b) Natural Language Understading (NLU) c) Natural Language Generation (NLG) d) Text to Speech (TTS)	L4	CO-03